

Chapter 2

Project Management & Information Technology

Learning Objectives

- Describe the systems view of project management and how it applies to information technology (IT) projects
- Understand organizations, including the four frames, organizational structures, and organizational culture
- Explain why stakeholder management and top management commitment are critical for a project's success

Learning Objectives

- Understand the concept of a project phase and the project life cycle, and distinguish between project development and product development
- Discuss the unique attributes and diverse nature of IT projects
- Describe recent trends affecting IT project management, including globalization, outsourcing, virtual teams, and agile project management

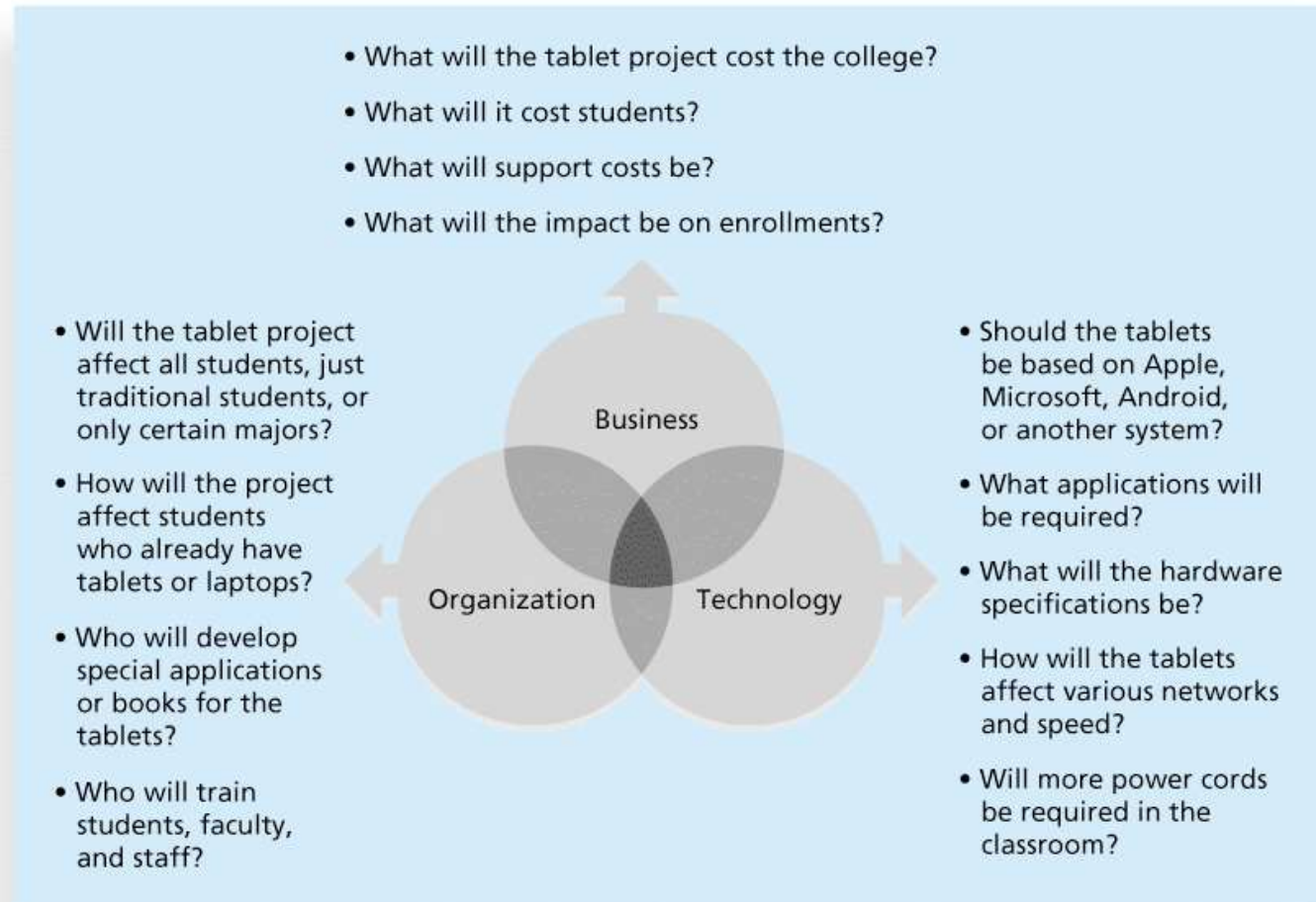
Projects Cannot Be Run In Isolation

- Projects must operate in a broad organizational environment
- Project managers need to use **systems thinking**
- Senior managers must make sure projects continue to support current business needs

A Systems View of Project Management

- A **systems approach** emerged in the 1950s to describe a more analytical approach to management and problem solving
- Includes:
 - **Systems philosophy**
 - **Systems analysis**
 - **Systems management**

Three Sphere Model for Systems Management



Perspectives on Organizations

Structural frame: Roles and responsibilities, coordination, and control. Organizational charts help describe this frame.	Human resources frame: Providing harmony between needs of the organization and needs of people.
Political frame: Coalitions composed of varied individuals and interest groups. Conflict and power are key issues.	Symbolic frame: Symbols and meanings related to events. Culture, language, traditions, and image are all parts of this frame.

What Went Wrong?

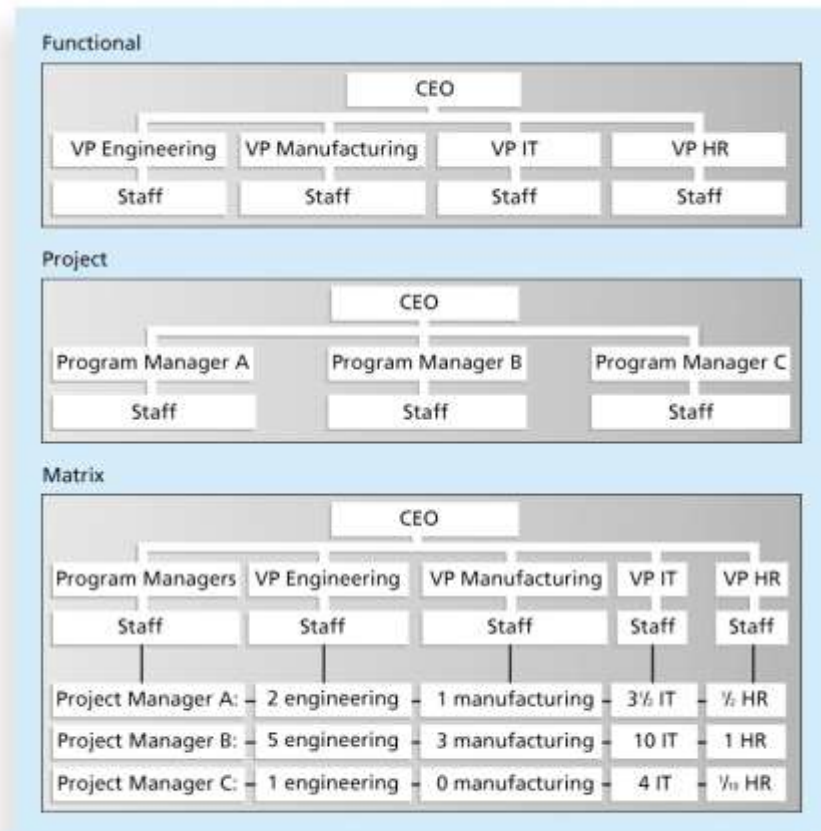
- In a paper titled “A Study in Project Failure,” two researchers examined the success and failure of 214 IT projects over an eight-year period in several European countries.
- The researchers found that only one in eight (12.5 percent) were considered successful in terms of meeting scope, time, and cost goals.
- The authors said that the culture within many organizations is often to blame
- Among other things, people often do not discuss important leadership, stakeholder, and risk management issues

Factors Influencing IT Project Management

- **Organizational Structure**
- **Organizational Culture**
- **Top Management Commitment**
- **Stakeholder Management**
- **Commitment to IT**
- **IT Governance/Organizational Standards**

Organizational Structures

- 3 basic organization structures
 - Functional
 - Project
 - Matrix



Organizational Structure Influences on Projects

Project Characteristics	Organizational Structure Type				
	Functional	Weak Matrix	Balanced Matrix	Strong Matrix	Project
Project manager's authority	Little or none	Limited	Low to moderate	Moderate to high	High to almost total
Percent of organization's personnel assigned full-time to project work	Virtually none	0–25%	15–60%	50–95%	85–100%
Who controls the project budget	Functional manager	Functional manager	Mixed	Project manager	Project manager
Project manager's role	Part-time	Part-time	Full-time	Full-time	Full-time
Common title for project manager's role	Project coordinator/ project leader	Project coordinator/ project leader	Project manager/ project officer	Project manager/ program manager	Project manager/ program manager
Project management administrative staff	Part-time	Part-time	Part-time	Full-time	Full-time

Organizational Culture

- **Organizational culture** is a set of shared assumptions, values, and behaviors that characterize the functioning of an organization

Characteristics of Org. Culture

- Member identity*
- Group emphasis*
- People focus
- Unit integration*
- Control
- Risk tolerance*
- Reward criteria*
- Conflict tolerance*
- Means-ends orientation
- Open-systems focus*

Top Management Commitment

- People in top management positions are key stakeholders in projects
- A very important factor in helping project managers successfully lead projects is the level of commitment and support they receive from top management

How does Top Management help a project?

Stakeholder Management

- Project managers must take time to identify, understand, and manage relationships with all project stakeholders
- Using the four frames of organizations can help meet stakeholder needs and expectations

Who is often considered one of the most important stakeholders?

Organizational Commitment to IT

- If the organization has a negative attitude toward IT, it will be difficult for an IT project to succeed
- Having a Chief Information Officer (CIO) at a high level in the organization helps IT projects
- Assigning non-IT people to IT projects also encourage more commitment

IT Governance / Standards

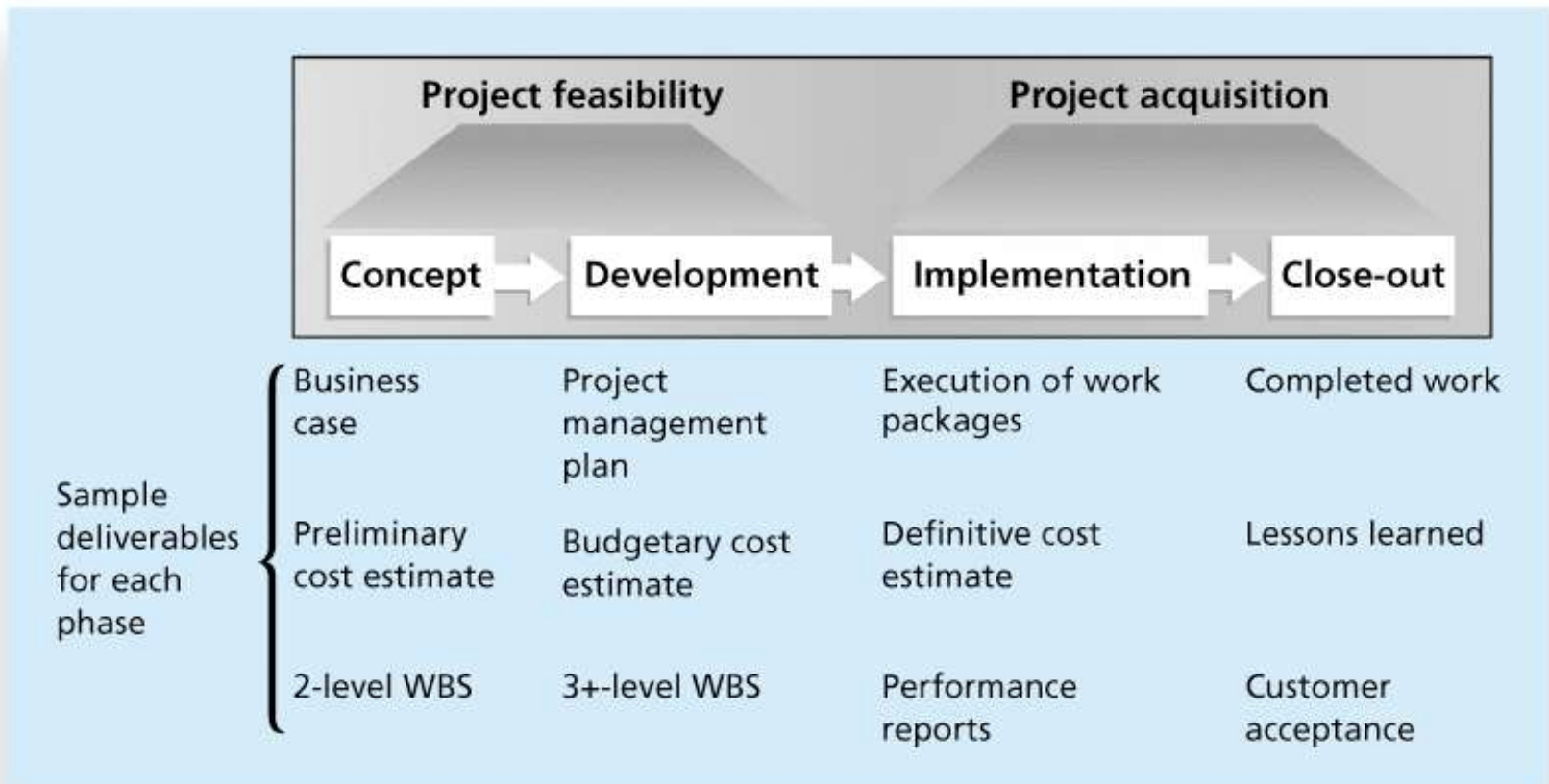
- **IT governance** addresses the authority and control for key IT activities in organizations, including IT infrastructure, IT use, and project management
- **Standards**
 - PMO or simply having templates/forms
 - Example: [Texas Tech IT PM Standards](#)

Project Phases and the Project Life Cycle

- A **project life cycle** is a collection of project phases that defines
 - what work will be performed in each phase
 - what deliverables will be produced and when
 - who is involved in each phase, and
 - how management will control and approve work produced in each phase

- A **deliverable** is a product or service produced or provided as part of a project

Phases of the Traditional Project Life Cycle

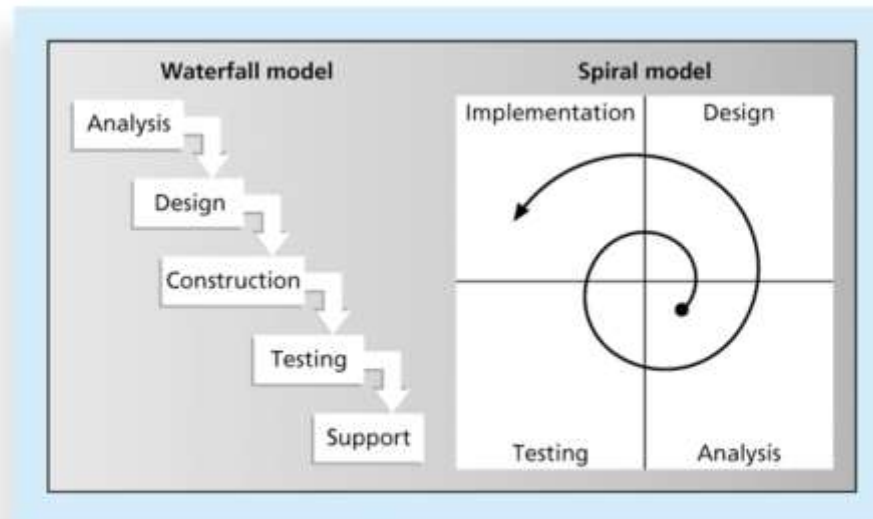


Product Life Cycles

- The **Systems Development Life Cycle (SDLC)** is a framework for describing the phases involved in developing and maintaining information systems
- Systems development projects can follow
 - **Predictive life cycle**
 - **Adaptive Software Development (ASD) life cycle**

Examples of Predictive Life Cycle Models

- Waterfall model
- Spiral model
- Incremental build model
- Prototyping model
- Rapid Application Development (RAD) model



Agile Software Development

- Agile software development has become popular to describe new approaches that focus on close collaboration between programming teams and business experts

The Importance of Project Phases and Management Reviews

- A project should successfully pass through each of the project phases in order to continue on to the next
- Management reviews, also called **phase exits** or **kill points**, should occur after each phase to evaluate the project's progress, likely success, and continued compatibility with organizational goals

What Went Right?

"The real improvement that I saw was in our ability to—in the words of Thomas Edison—know when to stop beating a dead horse....Edison's key to success was that he failed fairly often; but as he said, ***he could recognize a dead horse before it started to smell...In information technology we ride dead horses—failing projects—a long time before we give up.*** But what we are seeing now is that we are able to get off them; able to reduce cost overrun and time overrun. That's where the major impact came on the success rate."*

Many organizations, like Huntington Bancshares, Inc., use an **executive steering committee** to help keep projects on track.

*Cabanis, Jeannette, "'A Major Impact': The Standish Group's Jim Johnson On Project Management and IT Project Success," PM Network, PMI, Sep.1998, p. 7

The Context of IT Projects

- Diverse in terms of size, complexity, products produced, application area, and resource requirements
- Diverse IT Project Team Members
- Diverse Technologies

Article: Tech Project and Direction

- What have they been doing due to the recession and what direction are they moving into?
- What are some of the top projects?
- What industries are seeing growth in major IT projects?

Article: Project Management & Software Development

- Software development projects have distinct characteristics not found in other projects
- Based on the article, what are some of these characteristics?
- Which did you feel was the number one mistake a PM could do in software development projects?

Improving Likelihood of Success for IT Projects

- Value-Driven Approach
- Socio-Technical Approach
- Commit to Project Management Principles (Chaos Report)
- Knowledge-Management Approach

Recent Trends Affecting IT Project Management

- Globalization
- Outsourcing
- Virtual teams

Important Issues and Suggestions Related to Globalization

■ Issues

- Communications
- Trust
- Common work practices
- Tools

■ Suggestions

- Employ greater project discipline
- Think global but act local
- Keep project momentum going
- Use newer tools and technology

Outsourcing

- Organizations remain competitive by using outsourcing to their advantage, such as finding ways to reduce costs
- Their next challenge is to make strategic IT investments with outsourcing by improving their enterprise architecture to ensure that IT infrastructure and business processes are integrated and standardized

Virtual Teams: Advantages/Disadvantages

■ Advantages:

- Increasing competitiveness and responsiveness
- Lowering costs
- Increased expertise and flexibility
- Increased work/life balance

■ Disadvantages:

- Isolation & Interpersonal Relationships
- Communication Problems
- Dependence on Technology

Approaches and Philosophy

- eXtreme Project Management (XPM)
 - Holistic view of planning and managing
 - Projects don't always fit traditional scheduling and formal techniques (they are chaotic)
 - Focuses on the “human side” of PM

- Agile Project Management
 - Iterative, incremental approach
 - Popular
 - <http://www.pmi.org/Certification/New-PMI-Agile-Certification.aspx>

Chapter Summary

- Systems approach when working on projects
- Organization frames: structural, human resources, political, and symbolic
- Project managers need to consider several factors due to the unique context of information technology projects
- Recent trends affecting IT project management include globalization, outsourcing, virtual teams, and Agile